

David Jiang

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EDUCATION

University of Waterloo | *Honours Computer Science (co-op)*

Expected 2028

GPA: 3.96/4.0

- **Relevant coursework:** Data Structures & Data Management, Algorithm Design, Object-Oriented Programming, Compiler Design, Computer Hardware Architecture
- **Awards:** Canadian Computing Competition 95th percentile, Euclid Math Contest Distinction

EXPERIENCE

Full Stack Software Developer Intern

Jan. 2026 – Apr. 2026

APi Group Inc. (Habtech)

Mississauga, ON

- Built an in-house quotation platform (Next.js + .NET Core) serving 500+ users, **cutting ~\$2M/yr** in SaaS licensing
- Reduced bulk-import DB writes **85% (10k+ records)** by diffing rows against existing state and writing changed rows
- Engineered a no-code workflow engine backed by JSON-defined steps in a state table, letting non-technical staff configure multi-step approval flows and cutting 20+ eng requests/month
- Developed multi-tenant RBAC on a shared app and database, isolating client data across tenants while enforcing role-based permissions per request, scoping each user to their tenant and rank without per-client deployments

Software Engineering Intern

May 2025 – Aug. 2025

StackAdapt

Toronto, ON

- Automated cleanup across 100k+ entries, **reclaiming 90%+ of DB storage** and cutting latency from 10s+ to < 1s
- Built a multi-step campaign-creation wizard with centralized RTK Query caching, eliminating redundant re-fetches of large config datasets, cutting thousands of daily backend queries
- Cut Sentry error volume 40%+ by filtering noise, adding retry-with-delay logic for post-deletion DB sync failures
- Migrated ad-vetting to a service with auto-approval, deprecating all legacy logic and running DB migrations to clear stale data, cutting manual review ~20% for ad quality reviewers

PROJECTS

Camera Obstruction & Movement Detection App | *OpenCV, FastAPI, Next.js, PostgreSQL*

- Built a full-stack CV dashboard that flags camera tampering by scoring live frames against reference images with SSIM, ORB feature matching, and RANSAC homography inlier counts
- Developed an auto-tuning service that searches detection thresholds against labeled data to maximize F1, reaching **0.94 F1 on 500 samples** vs pre-picked thresholds
- Engineered a synthetic data generator to solve real-sample scarcity, producing training data without manual captures

Gameboy CPU Emulator | *C++, SDL2*

- Built a Game Boy CPU emulator in C++ implementing the full instruction set, with register/flag handling, memory-mapped addressing, and a fetch-decode-execute loop
- Implemented cartridge loading and an SDL2 output pipeline, decoding headers and memory-mapping ROM banks

Full Stack Task Planning Web Application | *SvelteKit, Firebase, Google Cloud Platform*

- Built a full-stack task app that prioritizes tasks on an Eisenhower matrix (importance vs. urgency), with real-time multi-device sync via Firebase Auth and Realtime Database
- Added a guest mode backed by in-memory Svelte stores, letting users try the app without creating an account

Python Social Media Short-Form Content Generator | *Python, Whisper, FFmpeg*

- Built an automated pipeline that overlays AI-generated narration and synced captions onto existing footage
- Leveraged the Whisper API to align captions to audio, compositing text and video into clips with FFmpeg/MoviePy

SKILLS

Languages: TypeScript, Python, JavaScript, C#, C++, C, SQL, Ruby, Java

Frameworks & APIs: React, Next.js, Node.js, .NET Core, FastAPI, Ruby on Rails, Redux, RTK Query, GraphQL

Tools & Cloud: Git, Docker, Azure (Web Apps, Pipelines, AD), PostgreSQL, SQL Server, Google Cloud, Linux, OpenCV